Problem 6.1.3. Dynkin diagrams. Let Γ be a graph, i.e., a finite set of points (vertices) connected with a certain number of edges (we allow multiple edges). We assume that Γ is connected (any vertex can be connected to any other by a path of edges) and has no self-loops (edges from a vertex to itself). Suppose the vertices of Γ are labeled by integers $1, \ldots, n$. Then one can assign to Γ an $n \times n$ matrix $R = (r_{ij})$, where r_{ij} is the number of edges connecting vertices i and j. This matrix is obviously symmetric and is called the **adjacency matrix**. Define the matrix $A = 2I - R$, where I is the identity matrix.

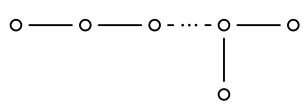
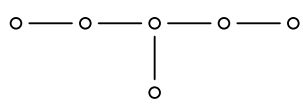
Definition 6.1.4. Γ is said to be a **Dynkin diagram** if the quadratic form on $\mathbb{R}^n$ with matrix A is positive definite.

Dynkin diagrams appear in many areas of mathematics (singularity theory, Lie algebras, representation theory, algebraic geometry, mathematical physics, etc.). In this problem you will get a complete classification of Dynkin diagrams. Namely, you will prove

Theorem. Γ is a Dynkin diagram if and only if it is one of the following graphs:

- A_n :

- D_n: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